

LIGHTNING DETECTION SYSTEM

Technical Data Sheet



1. Product Overview

A standalone, solar-powered lightning detection system designed for industrial sites, weather stations, and remote installations. The system provides continuous autonomous monitoring with real-time cloud data publishing, proximity-based alerts, and comprehensive local data logging.

Engineered for reliable operation in demanding electromagnetic environments where conventional detectors may trigger false alarms due to nearby electrical equipment such as arc furnaces, variable-frequency drives, and switchgear.

Key Features

- Continuous 24/7 detection up to 40 km range
- 14-level storm distance estimation
- Cloud-to-ground and intra-cloud detection
- Configurable EMI rejection for high-noise sites
- Solar-powered with lithium-ion battery backup
- Real-time cloud dashboard via HTTPS (TLS 1.2+)
- Configurable proximity alerts (SMS, WhatsApp, email)
- Automated periodic reports with PDF attachment
- IP67 weatherproof non-metallic enclosure
- Offline data buffering with automatic cloud sync

2. Technical Specifications

2.1 Detection Performance

Parameter	Value
Maximum Detection Range	40 km
Distance Estimation	14 discrete levels + out of range
Reported Distances (km)	1, 5, 6, 8, 10, 12, 14, 17, 20, 24, 27, 31, 34, 37, 40
Distance Accuracy	±4 km (manufacturer rated)
Detection Types	Cloud-to-ground, intra-cloud
Lightning Energy	21-bit relative measurement per event
Antenna Frequency	500 kHz (±3.5%)
Noise Floor Levels	0–7 (programmable)
Min. Strike Threshold	1, 5, 9, or 16 events
Event Latency	< 2 seconds (detection to log)

2.2 System Performance

Parameter	Value
Controller	Quad-core ARM Cortex-A53 SBC
Sensor Interface	SPI Mode 1, 1 MHz
Connectivity	2.4 GHz 802.11 b/g/n Wi-Fi
Cloud Protocol	HTTPS POST (TLS 1.2+, JSON)
Local Logging	CSV, daily rotation at midnight UTC
Time Synchronisation	NTP (UTC)
Report Delivery	Email with PDF (configurable period)

2.3 EMI Rejection

Built-in disturber rejection filters interference from arc furnaces, variable-frequency drives, crane motors, bus bars, switchgear, and welding equipment. Parameters are fully configurable and optimised during on-site commissioning.

3. Communication and Alerts

3.1 Cloud Data Publishing

Parameter	Value
Protocol	HTTPS POST (JSON payload)
Encryption	TLS 1.2+ end-to-end
Authentication	API key per station
Dashboard	Real-time web-based visualisation

3.2 Proximity Alerts

Parameter	Value
Default Alert Distance	10 km (configurable 1–40 km)
Cooldown Period	60 minutes (configurable)
Channels	SMS, WhatsApp, Email
Multiple Recipients	Yes (configurable per channel)
Content	Distance, energy, timestamp, station ID

3.3 Periodic Summary Reports

Automated lightning reports distributed via email at a configurable interval (weekly or monthly). Reports include total events, closest approach, energy statistics, system uptime, and a formatted PDF attachment.

4. Power System

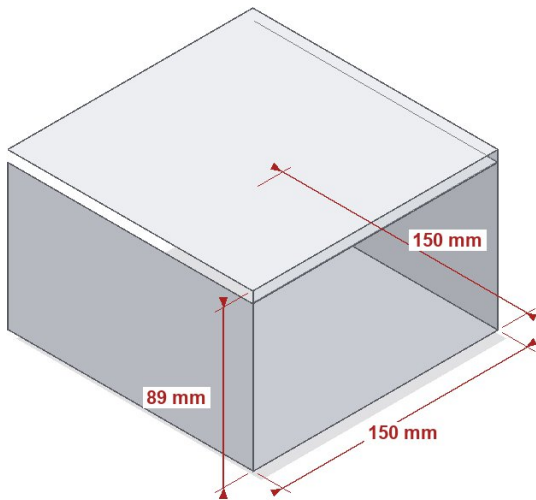
Parameter	Value
Primary Source	Solar panel (6–24 V DC input)
Charge Controller	MPPT (Maximum Power Point Tracking)
Battery	3× 18650 Li-ion parallel
Output	5 V regulated, 3 A max
Protection	Over-charge, over-discharge, over-temperature

Minimum recommended panel: 5 W. System is energy-positive with 4–6 peak sun hours per day. Compatible with existing solar infrastructure or a dedicated panel.

5. Enclosure

5.1 IP67 Enclosure

Parameter	Value
Material	ABS plastic (non-metallic, RF transparent)
External Dimensions	150 × 150 × 89 mm (5.9" × 5.9" × 3.5")
Ingress Protection	IP67 (dust-tight, short-term submersion)
Seal	Silicone gasket
Closure	Stainless steel hinged latches
Cable Entry	PG9 IP68-rated cable glands (bottom entry)
Mounting	Integrated brackets + internal mounting plate
Mast Fit	30–50 mm diameter (U-bolt or clamp)



6. Installation

6.1 Site Requirements

- Weather station mast or mounting pole (30–50 mm diameter)
- Solar panel (6–24 V, 5 W minimum) or existing infrastructure
- 2.4 GHz Wi-Fi coverage at installation point
- Clear sky exposure (no overhead metallic structures above sensor)

6.2 Commissioning

Professional installation and commissioning includes:

- Physical mounting and cable entry
- Antenna calibration and frequency verification
- EMI noise profiling and filter optimisation
- Cloud platform and alert configuration
- Training for on-site personnel

6.3 Network Requirements

Service	Protocol	Port
Cloud publishing	HTTPS	443 (outbound)
Time sync	NTP	123 (outbound)
Email reports	SMTP/TLS	587 (outbound)
Alert APIs	HTTPS	443 (outbound)